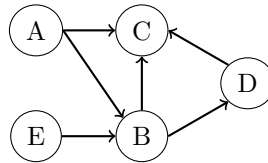


Worksheet 3

Exercise 1: D-Separation

Have a look at the following causal graphical model from the worksheet 2.



Check the following d-Separations.

1. Are E and D d-separated by B?
2. Are E and C d-separated by A?

1. Are E and D d-separated by B?

We need to make sure that all paths from E to D are blocked by conditioning on B. Conditioning on B is blocking the chain from $E \rightarrow B \rightarrow D$, the chain from $E \rightarrow B \rightarrow C$ and opening the collider from $E \leftarrow B \rightarrow A$. Therefore, all paths via the two chains are blocked. The remaining two paths $E \leftarrow B \rightarrow A \rightarrow C \rightarrow D$ and $E \leftarrow B \rightarrow A \rightarrow C \rightarrow B \rightarrow D$ contain both another collider in C which is blocking. Therefore, E and D are d-separated by B.

2. Are E and C d-separated by A?

There are three paths from E to C which need to be blocked: (E,B,A,C), (E,B,C) and (E,B,D,C). Since there is a chain ($E \rightarrow B \rightarrow C$), we have an open path from E to C. This path is also open, when we condition on A. Therefore, E and C are **not** d-separated by A.

We can check the answers with R:

```

library(gRbase)
# List all nodes with their parents
graphx <- dag("A",
              c("B", "A", "E"),
              c("C", "A", "B", "D"),
              c("D", "B"),
              c("E"))
d.separates <- function(a,b,c,dag){
  separates(a,b,c, moralize(ancestralGraph(union(union(a,b),c),dag)))}
d.separates("E", "D", "B", graphx)

```

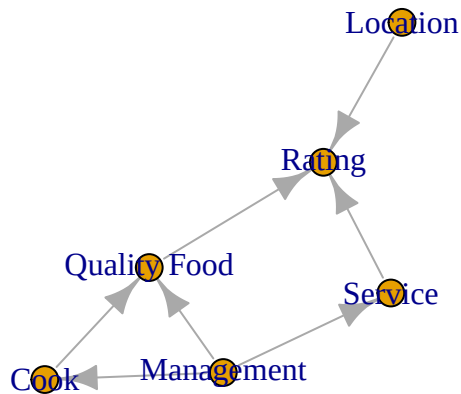
```
[1] TRUE
```

```
d.separates("E", "C", "A", graphx)
```

```
[1] FALSE
```

Exercise 2: Rating of a restaurant

The following causal graph is showing factors that influence the overall rating of a restaurant.



a) How does this graph look like if we plan an intervention on quality of food?

```

# Original Graph:
library(gRbase)
grest <- dag(c("QualityFood", "Management", "Cook"),
             c("Cook", "Management"),
             c("Service", "Management"),
             c("Rating", "QualityFood", "Service", "Location"))
plot(grest)

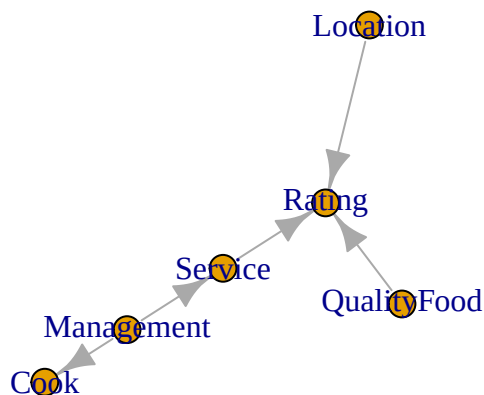
```

Intervention on quality of food means that all edges between the node quality of food and its parents are removed.

```

# Entfernen der Kanten zu den Eltern:
grest_int <- removeEdge("Management", "QualityFood", grest)
grest_int <- removeEdge("Cook", "QualityFood", grest_int)
# Interventionsgraph:
plot(grest_int)

```



b) List all possible adjustment set given by the backdoor criterion to estimate the causal effect of the cook on the rating?

To estimate the causal effect we need to block all backdoor paths from cook. There are 2 backdoor paths:

- 1) Cook $\leftarrow$ Management $\rightarrow$ Quality of Food $\rightarrow$ Rating
- 2) Cook $\leftarrow$ Management $\rightarrow$ Service $\rightarrow$ Rating

Path 1 is blocked by Management and/or Quality of Food.
Path 2 is blocked by Management and/or Service.

Possible sets for d-Separation are:

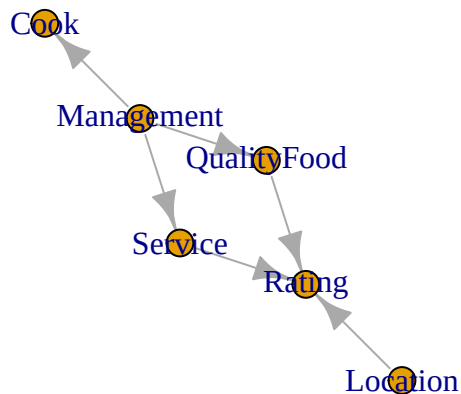
{Management}, {Management, Service}, {Management, QualityFood}, {Management, QualityFood, Service}, {QualityFood, Service}

However, we need to remove all nodes from the sets that are descendants of cook (i.e. the node is on the causal path from cook to rating). Therefore, we obtain only the following adjustment set:

{Management}, {Management, Service}

The backdoor criterion is related to d-Separation as follows: If a set Z satisfies the backdoor criterion relative to X and Y, then X and Y are d-separated by Z in the manipulated causal graph, which is the causal graph obtained by removing all arrows starting from X. Therefore, we can test the adjustment sets for validity by:

```
# Manipulated Graph
grest_cook <- removeEdge("Cook","QualityFood",grest)
plot(grest_cook)
```



```
# Function to test d-separation
library(gRain)
d.separates <- function(a,b,c,dag){
  separates(a,b,c, moralize(ancestralGraph(union(union(a,b),c),dag)))}
# Checking the d-separations in the manipulated (!!) graph
# Testing Z = {Management}
d.separates("Cook","Rating", "Management", grest_cook)
```

```
[1] TRUE
```

```
# Testing Z = {Service, Management}
d.separates("Cook","Rating",c("Management","Service"), grest_cook)
```

```
[1] TRUE
```

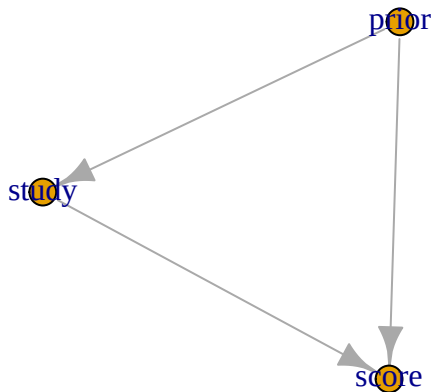
Exercise 3: Exam Score

Imagine we want to analyze the effect of study hours on exam scores. However, a student's prior knowledge influences both the amount of time spent studying and the exam performance. The data of this study are available in the data.frame `exam`, which is stored in the file `exam.rda`.

```
load("exam.rda")
```

a) Draw the causal diagram.

```
library(gRbase)
# List all nodes with their parents
graph.exam <- dag("prior",
                  c("study", "prior"),
                  c("score", "study", "prior"))
# Draw graph
plot(graph.exam)
```



b) What is the probability of getting a good grade on the exam?

```
library(gRain)
# combine graph and data to a Bayesian network
gn <- grain(graph.exam, data = exam)
# marginal probabilities
querygrain(gn, nodes = "score", type = "marginal")
```

```
$score
score
      high      low
0.6480118 0.3519882
```

The probability to have a high score is 64.8 %.

An alternative way to calculate this is:

The probability can be estimated by calculating the proportion of students with good exam grades out of the total of 679 students observed.

```
sum(exam$score == "high")/nrow(exam)
```

```
[1] 0.6480118
```

c) Calculate the conditional probability of a high score given that a student studies much, i.e. $P(\text{score} = \text{high} \mid \text{study} = \text{much})$, and the conditional probability of a high score given

that a student studies little, i.e. $P(\text{score} = \text{high} \mid \text{study} = \text{little})$.

```
# extract conditional probabilities from gn
querygrain(gn, nodes = c("score", "study"), type = "conditional")
```

```
      study
score  little      much
high 0.3097015 0.8686131
low  0.6902985 0.1313869
```

The conditional probabilities are

$P(\text{score} = \text{high} \mid \text{study} = \text{much}) = 0.8686$ and $P(\text{score} = \text{high} \mid \text{study} = \text{little}) = 0.3097$.

An alternative way to calculate this is:

To calculate the conditional probability, let's first consider only the subset of students with a lot of study time and then only those with little study time. For each of these two subsets, we calculate the proportion of students with good exam grades:

```
sum(exam$score == "high" & exam$study == "much") / sum(exam$study == "much")
```

```
[1] 0.8686131
```

```
sum(exam$score == "high" & exam$study == "little") / sum(exam$study == "little")
```

```
[1] 0.3097015
```

- d) Calculate the probability of a high exam score under the hypothetical intervention that all students have studied much, i.e. $P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{much}))$.

In order to calculate $P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{much}))$, we need to adjust for the prior knowledge (parental adjustment):

$$P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{much})) = \sum_{x=\text{little}/\text{much}} P(\text{score} = \text{high} \mid \text{study} = \text{much}, \text{prior} = x) \cdot P(\text{prior} = x)$$

```
# P(prior = little)
p1 <- querygrain(gn, nodes = "prior", type = "marginal")
p1l <- p1$prior["little"]
# or p1l <- sum(exam$prior == "little")/nrow(exam)
# P(prior = much)
p1m <- p1$prior["much"]
# or p1m <- sum(exam$prior == "much")/nrow(exam)

# P(score = high | study = much, prior = little)
p2 <- querygrain(gn, nodes = c("score", "study", "prior"), type = "conditional")
p2l <- p2["high", "much", "little"]
# or
# exam_sub_l <- exam[exam$study == "much" & exam$prior == "little",]
# p2l <- sum(exam_sub_l$score == "high") / nrow(exam_sub_l)

# P(score = high | study = much, prior = much)
p2m <- p2["high", "much", "much"]
# or
# exam_sub_m <- exam[exam$study == "much" & exam$prior == "much",]
# p2m <- sum(exam_sub_m$score == "high") / nrow(exam_sub_m)
```

```
# P(score = high | do(study = much))
p_im <- p2l * p1l + p2m * p1m
p_im
```

```
little
0.8815087
```

$P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{much})) = 0.88$. If we force all students to study a lot, then the probability to have a high score is 88%.

Remark:

$P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{much})) = 0.88 \neq 0.87 = P(\text{score} = \text{high} \mid \text{study} = \text{much})$

- e) Calculate the probability of a high exam score under the hypothetical intervention that all students have studied little, i.e. $P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{little}))$.

The interventional probability $P(\text{Score} = \text{high} \mid \text{do}(\text{Study} = \text{little}))$ can be calculated in the same way as in subtask d):

```
# P(score = high | study = little, prior = little)
p3 <- querygrain(gn, nodes = c("score", "study", "prior"), type = "conditional")
p3l <- p3["high", "little", "little"]
# or
# exam_sub_l <- exam[exam$study == "little" & exam$prior == "little",]
# p3l <- sum(exam_sub_l$score == "high") / nrow(exam_sub_l)
# P(score = high | study = little, prior = much)
p3m <- p3["high", "little", "much"]
# or
# exam_sub_m <- exam[exam$study == "little" & exam$prior == "much",]
# p3m <- sum(exam_sub_m$score == "high") / nrow(exam_sub_m)
# P(Score = high | do(Study = little))
p_il <- p3l * p1l + p3m * p1m
p_il
```

```
little
0.2340055
```

$P(\text{Score} = \text{high} \mid \text{do}(\text{Study} = \text{little})) = 0.23$. If we force all students to study little, then the probability to have a high score is 23%.

- f) What can you say about the effect of learning on the exam grade, taking into account the calculations from the previous subtasks?

There is a clear positive causal effect of studying causing a higher exam score:

$\text{ACE} = P(\text{Score} = \text{high} \mid \text{do}(\text{Study} = \text{much})) - P(\text{Score} = \text{high} \mid \text{do}(\text{Study} = \text{little}))$

```
p_im - p_il
```

```
little
0.6475032
```